

Required Command Lines and Installations for Running the Project Code

Best setup

Because this project uses `llama-cpp-python`, `faiss`, `mediapipe`, `fer`, and `sentence-transformers`, the safest route is a fresh Conda environment with Python 3.11. `llama-cpp-python` supports Python 3.8+ and can use prebuilt CPU wheels; `fer` also requires Python 3.8+; MediaPipe Face Landmarker needs a separate local `face\_landmarker.task` model file.

Required packages

- `numpy`
- `pandas`
- `openpyxl`
- `opencv-python`
- `mediapipe`
- `fer`
- `faiss-cpu`
- `sentence-transformers`
- `llama-cpp-python`
- `huggingface-hub` if you use `Llama.from\_pretrained(...)` download logic.

Required files

- `face\_landmarker.task`
- one local `.gguf` model file, such as `tinyllama-1.1b-chat-v0.3.Q4\_0.gguf` or whichever GGUF your code points to.

Create and activate the environment

```
conda create -n quizenv python=3.11 -y
conda activate quizenv
python -m pip install --upgrade pip setuptools wheel
```

Install all Python packages required by your code

```
python -m pip install numpy pandas openpyxl opencv-python mediapipe fer faiss-cpu sentence-
transformers huggingface-hub
python -m pip install llama-cpp-python --extra-index-url https://abetlen.github.io/llama-cpp-
python/whl/cpu
```

Why these

- `opencv-python`` provides OpenCV wheels.
- `fer`` is your emotion-recognition package.
- `faiss-cpu`` is the PyPI package for `import faiss``.
- `sentence-transformers`` provides `SentenceTransformer``.
- `openpyxl`` is needed for `DataFrame.to_excel(...)`` with `.xlsx``.
- `llama-cpp-python`` is the package behind `from llama_cpp import Llama``; it can install from source or use a prebuilt CPU wheel.

Put the required model files beside your script

```
E:\PythonProject\dynamic.py
E:\PythonProject\face_landmarker.task
E:\PythonProject\tinyllama-1.1b-chat-v0.3.Q4_0.gguf
```

Why

- MediaPipe Face Landmarker requires a downloaded local `.task`` model file and expects its path in `model_asset_path``.
- `llama-cpp-python`` expects `model_path`` to point to a real local `.gguf`` model file unless you use its Hugging Face download helper.

Test each important dependency first

```
python -c "import cv2; print('opencv ok')"
python -c "import mediapipe as mp; print('mediapipe ok')"
python -c "from fer.fer import FER; print('fer ok')"
python -c "import faiss; print('faiss works')"
python -c "from sentence_transformers import SentenceTransformer; print('sentence-transformers ok')"
python -c "from llama_cpp import Llama; print('llama-cpp ok')"
python -c "import pandas as pd; import openpyxl; print('pandas/openpyxl ok')"
```

Full install block

```
conda create -n quizenv python=3.11 -y
conda activate quizenv
python -m pip install --upgrade pip setuptools wheel
python -m pip install numpy pandas openpyxl opencv-python mediapipe fer faiss-cpu sentence-transformers huggingface-hub
python -m pip install llama-cpp-python --extra-index-url https://abetlen.github.io/llama-cpp-python/whl/cpu
python E:\PythonProject\dynamic.py
```